

CHATTANOOGA, TN, USA

WMO: 723240

Lat: 35.034N

Lon: 85.200W

Elev: 670

StdP: 14.34

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 13882

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	19.2	23.4	4.2	7.0	26.1	8.3	8.6	29.4	20.7	42.0	18.7	41.3	7.4	330	0.389

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.2	94.8	73.9	92.6	73.8	90.6	73.4	77.5	88.3	76.6	86.9	75.7	85.4	6.9	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
74.7	133.0	81.3	73.7	128.6	80.4	72.9	125.0	79.9	41.3	88.5	40.3	87.1	39.5	85.8	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.0	16.1	13.6	13.4	97.2	5.3	3.4	9.6	99.7	6.5	101.7	3.5	103.6	-0.3	106.0		
(5)			11.4	79.2	4.9	1.2	7.9	80.1	5.0	80.7	2.3	81.4	-1.2	82.2	(4)	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	61.8	41.3	45.4	53.1	61.9	69.9	77.2	80.1	79.8	73.8	62.7	51.3	44.5
	DBStd	15.60	10.02	9.27	8.82	7.47	6.28	4.40	3.42	3.79	5.67	7.65	8.03	8.88
	HDD50	869	305	181	71	6	0	0	0	0	0	5	80	221
	HDD65	3037	736	551	379	145	30	0	0	0	7	135	416	638
	CDD50	5179	35	51	166	362	615	817	932	923	714	397	118	49
	CDD65	1872	0	1	10	51	180	368	467	458	270	62	4	1
	CDH74	17363	0	6	98	526	1608	3417	4557	4398	2230	494	28	1
	CDH80	7052	0	0	11	105	512	1451	2018	1975	873	105	2	0
Wind	WSAvg	4.9	5.8	6.0	6.4	5.9	4.8	4.3	4.1	3.8	3.8	4.2	4.6	5.2
Precipitation	PrecAvg	53.6	4.9	4.6	5.6	4.5	4.1	3.9	5.0	3.5	4.3	3.3	4.7	5.2
	PrecMax	73.7	9.8	9.7	16.3	11.9	11.2	9.2	11.9	7.6	14.2	7.2	9.6	13.7
	PrecMin	35.1	0.9	0.7	1.2	0.7	1.1	0.6	0.8	0.0	0.5	0.1	1.5	0.9
	PrecStd	10.0	2.1	2.3	3.2	2.5	2.1	2.0	2.7	1.6	3.1	2.0	1.9	2.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	69.6	74.3	81.4	86.5	90.7	95.7	97.1	98.2	95.8	87.1	78.1	71.9
		MCWB	61.1	62.0	63.3	66.6	71.1	73.6	75.0	74.9	70.5	69.1	63.9	62.7
	2%	DB	65.4	69.3	76.7	83.1	88.0	92.9	93.9	94.9	91.9	82.7	72.8	67.1
		MCWB	58.2	58.8	61.3	64.8	69.8	73.6	75.2	74.6	70.9	67.3	60.4	60.8
	5%	DB	61.6	65.2	72.8	80.1	85.6	90.5	91.9	92.1	88.5	79.5	69.1	62.7
		MCWB	55.5	55.1	58.4	63.4	69.2	72.9	74.9	74.0	70.9	65.6	59.5	57.0
	10%	DB	57.3	60.9	68.6	76.7	82.8	88.0	89.6	89.6	85.2	76.1	65.4	58.9
		MCWB	51.7	52.6	56.5	62.0	68.4	72.1	74.4	73.6	70.4	64.4	57.5	53.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	63.5	64.7	66.3	70.8	74.5	77.4	79.2	78.7	75.4	73.1	67.9	66.0
		MCDB	66.9	70.9	75.7	80.6	84.5	88.8	89.7	90.3	84.9	81.7	71.7	69.6
	2%	WB	60.4	61.8	63.9	68.3	72.8	76.2	77.7	77.5	74.3	70.9	64.7	61.8
		MCDB	64.7	67.7	72.8	77.9	83.1	87.0	88.2	88.7	83.5	78.7	69.5	65.7
	5%	WB	56.8	58.1	61.6	66.3	71.4	75.2	76.8	76.4	73.3	68.9	62.1	58.5
		MCDB	60.7	62.6	69.2	75.3	81.2	85.5	87.0	86.7	82.2	75.5	67.2	61.6
	10%	WB	52.7	54.2	59.1	64.4	70.1	74.2	75.9	75.4	72.3	66.9	58.6	54.6
		MCDB	55.9	59.2	66.2	72.9	79.0	83.8	85.6	84.8	80.5	73.4	63.8	58.2
Mean Daily Temperature Range	5% DB	MDBR	17.8	19.0	20.9	22.5	20.8	19.3	18.2	18.7	19.5	21.2	20.8	17.3
		MCDBR	21.4	25.0	27.0	26.7	24.1	22.5	21.1	22.1	24.1	24.4	25.8	21.0
		MCWBR	15.0	15.4	13.3	11.4	8.7	7.1	6.2	6.2	7.4	10.6	14.7	14.5
	5% WB	MCDBR	17.8	19.8	21.3	20.8	19.7	19.0	18.4	18.5	17.7	17.7	18.5	17.4
		MCWBR	14.1	15.0	12.7	10.6	8.4	7.1	6.3	6.3	6.9	9.2	13.0	13.9
Clear-Sky Solar Irradiance	taub	0.322	0.334	0.364	0.411	0.465	0.504	0.549	0.550	0.438	0.380	0.343	0.318	
	taud	2.486	2.448	2.395	2.250	2.136	2.065	1.960	1.939	2.241	2.395	2.442	2.518	
	Ebn at Noon	280	289	288	278	263	252	239	236	260	268	268	274	
	Edh at Noon	27	31	36	43	49	53	58	58	41	32	28	25	
All-Sky Solar Radiation	RadAvg	762	931	1296	1672	1846	1957	1853	1730	1484	1208	876	651	
	RadStd	57	119	121	111	141	142	116	91	138	154	100	65	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	+0.99	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+302	+216
(47) Regional (34 neighbors)	+0.85	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+259	+185

Nomenclature: See separate page